WILD WEATHER

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ABSTRACT

My artist talk presents the prototype for the installation of a networked public sound sculpture planned for the Newcastle central business district in late 2011. The project is supported by the Newcastle City Centre Committee who, along with the organisation LiveSites, are seeking to install a range of projects over the coming year and are actively exploring strategies for interactive and innovative projects in the public domain.

THE INSTALLATION

The Wild Weather public sound installation includes five light weight transparent acrylic resonators, (approx 2200mm high X 110mm in outside diameter) suspended side by side and mechanically controlled to oscillate in response to the file transfer of media (predominantly audio & text files) from a remote server to the sculpture's interface (text files will be generated as audio files using text to speech software and the application WireTap Studio). I am currently working with both Arduino and Infusion Systems analog to MIDI digitisers and sensor technologies with standalone (using the Arduino programming language similar to C++) and Max/MSP applications on a MacBook Pro computer to determine the best solution. The options for this installation include using a remote computer streaming media from a server or having all media files transferred (using FTP) to local storage, but this is yet to be determined. The advantage of the local storage option is that all media will continue to be available for playback regardless of Internet connectivity. (See Figure 6).

I plan to use the FTP application Transmit to transfer and cue media files from the wildweather.net.au website to a dedicated server to the sculpture's hard drive. This can be done very effectively through an open source CMS Drupal website where individual or community based users can register and upload content (including a range of media files) for screening and subsequent broadcast, respond to specific calls for works and listen to and see the resonators in action through a webcam installed at the site. If the media uploaded is an audio file it would typically be in the stereo WAV or AIFF format, 16 bit/ 44.1Khz. Its duration would not exceed 4 to 5 minutes and so have an average file size of 40 to 50MB (10MB/ minute). To manage all of the server operations I plan to customise my own (dv) Dedicated-Virtual Server from the company Media Temple. This service is scalable (depending on usage) and would start with 20GB Storage, 512MB RAM and 1TB of Transfer Bandwidth, sufficient to manage and control my initial hosting environment.

The sculpture's sound projection is planned to emanate from each of the five resonators, equipped at their base with a small four inch speaker. The media content transferred via the Internet to the sculpture's computer would playback through any one of these five speakers. I am also experimenting with generating sound from within the five resonators using either prerecorded samples or natural sound. For example, the resonators could contain a quantity of liquid inside their closed end acrylic tubes. When set in motion the natural sound picked up (using very sensitive microphones) by the movement of the liquid inside the oscillating resonators could then be processed using a range of signal processing techniques (such as time stretching and delay) and then amplified back through their speakers. To accomplish this however the movement of liquid inside the resonators would have to be extreme and whether or not this is achievable is yet to be determined. The swing span for example of the resonators would have to be greater than 180 degrees to tip the liquid and the length of the resonators would need to be shorter than I have proposed.

The sculpture is designed as an Internet-orientated real-time interactive system for use by the general public. It is a hyperinstrument designed to be used with an analog to MIDI digitiser and five actuator signals with high resolution to transmit messages through USB to a computer where they are made available at a virtual serial port for analysis and/or control purposes. The research focus of the project is on designing a system (using software, sensors, signal processing and sculptural objects) that manages and broadcasts human response to human ecology, as well as exploring the appropriate protocols and innovative content for interactive art in public environments free of physical, social and/or political constraints. The Wild Weather installation can be installed in any public space with access to electricity, the Internet and an under cover public area with an approximate ceiling height of 4.5 meters, a minimum room width of 3.6 meters and a total swing span of 4.6 meters. The interaction with the sculpture will take place entirely over the Internet and is not constructed for any form of physical human contact.

Project Theme

We make the weather, we take the temperature, the temperature is rising, the weather is wilder. Professor of natural hazards at James Cook University, Cyclone

chaser Jonathan Nott this year warned that, *"town planners have become "very, very complacent" about the dangers of wild weather which is likely to increase during La Nina weather patterns and a shift in the Pacific Decadal Oscillation, which will warm the east Pacific ocean and possibly fuel severe cyclones. Three have hit Queensland's coast already this summer; Yasi, Anthony, and the weakest Tasha, which triggered the flooding rains that left three quarters of the state a disaster zone"* (Bita 2011).

This project addresses this intense and dangerous relationship we have with the weather and focuses on the devastating affects of floods and storms on our landscape and it's population. *"Learning to live with floods rather than fighting the rivers"*, where *"after 10 years of draught the floods have arrived on the Australian landscape"* are now commonly repeated messages in today's media. Central to this project is an investigation into these shared experiences and threatening relationships. Through the use of networked technologies the installation will provide the general population with a means to privately reflect on their experiences and relationships with their local weather environment and where possible share meaningful content that might describe the social and physical impact of weather on their communities. A diversity of stories for broadcast might explore their experiences, observations, memories and perceptions of weather change in different regions and may include flood stories, meteorological stories, cloud stories, rain, storm and hail stories and other observations of change to the weather. An extensive range of form in the content could include sound recordings, anecdotes, recollections, poetry, fragments, old recordings, letters, emails, blogs, lyrics and any other material that can be translated into sound. Prior to the public installation of the work an open call for content will take place and that addressing the theme will be made available for sculptural broadcast.

Project history

This work expands on and is inspired by my previous involvement in a exhibition titled Memory Flows curated by Norie Neumark, Deborah Turnbull and Sophia Kouyoumdjian in May/June 2010. Memory Flows is an ongoing and distributed media art project of the CMAI (Center for Media Arts Innovation) and was exhibited in collaboration with the Sydney Olympic Park Authority and UTS at the Armory Gallery, Newington Armory, Sydney. Norie Neumark explains the exhibition in her article, Memory Flows: Rivers, Creeks & the Great Artesian Basin in that, *"This exhibition works with the figure of memory to suggest that we are in an active and ethical relationship with our waterways. If we think of rivers as having memory - as an affordance we can connect to - we can sense their 'otherness' calling out to us, provoking desire and curiosity about our relations with such natural others. We can sense the flow of memory between ourselves and rivers as a network of complex relations"*. (Neumark 2010). For the Memory Flows exhibition composer/programmer Solange Kershaw and I installed a motion sensitive sound sculpture titled Memory Pendulums. Affected by contact and gravity the gentle swinging movement of the sculpture's pendulums modulated the spoken measure of a poem written for the work by the Australian poet Jill Jones. Her words flowed randomly in time with the playful swinging back and forth of three diamond shaped acrylic water pendulums, each filled to a different capacity with a blue (toxic) liquid, triggering a different memory, dripping into and polluting the muddy waters of the Parramatta River contained in a basin just below the tip of the pendulums. The three pendulums controlled the listening of the poem. If they swung uninterrupted for approximately 90 seconds the listener would hear the entire poem, which in three parts took all three pendulums to swing. The mechanism used to trigger the poem was created using a Max/MSP patch on a MacBook Pro, custom built sensors, mercury switches and the I-CubeX MicroDig.

Memory Pendulums Max/MSP Patch

The central object for this patch was the usb-microdig. The physical USB-microDig from Infusion Systems (Figure 1) is a thumb-sized hardware device that encodes analog signals to MIDI. It includes 8 digital outputs/ inputs that can be used to control actuators or trigger MIDI via sensors. The firmware for the USB-microDig enables it to operate in both stand-alone mode (sensor data is processed before it is transmitted) as well as host mode (raw sensor data is transmitted). The 8 inputs of each 10 bits resolution (1024 steps of each 4.9 mV) can sample at up to 5760 Hz with milliseconds latency to capture even the finest gestures and movements. (Mulder 2011).



Figure 1. I-CubeX USB-microDig

The Max patch was co-written by my project partner Solange Kershaw and myself. It is described as follows: For the Memory Flows exhibition we connected three custom built tilt sensors to the inputs of the USB-microDig. The sensor data was routed to the first three digital outputs and used to control the rest of the patch. The first input in the microdig object was used to determine the port, in this case a virtual serial (usb) port. The next 8 inputs were the sensor inputs in the physical microDig. These sensor inputs had to be turned on, and for the work to self-start, we used the 'loadbang' object. The loadbang object generates a 'bang' when a patcher is loaded, which in turn was connected to a toggle object (toggle object switches on and off, 1 and 0). We actually used a number of loadbangs delayed by 1 second to ensure that the digital outputs were reliably triggered, regardless of how fast or slow the computer, softwares and patch were loading on startup (Figure 2).

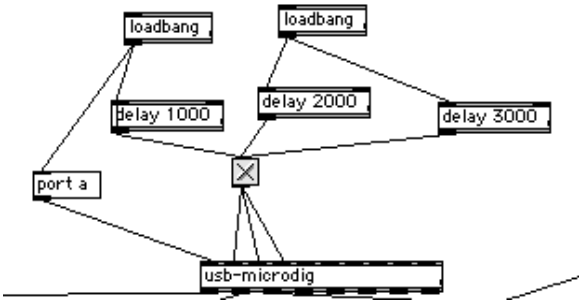


Figure 2. Max Patch loadbangs on startup

The first object connected to the digital output was a number box that displayed the sensor data in the midi range : 0 - 127. To determine whether the tilt switch was in the 'on' position (ie, when the sensor data ouput was 127) the number box was connected to a select object, that would trigger a bang in its output only if its input matched its argument. The select object here triggered two signal paths, one to play the sound file and one to monitor the time elapsed after that initial trigger (in 500ms units). As the pendulums swung through each cycle, the sensor data output changed from 127 to 0 and back again as the tilt switches went through each on/off state. So as to not restart the process and audio file playback on each cycle the right button was connected to a onebang object. The onebang object let a bang in through the left inlet only if a bang had been received in the right inlet (Figure 3).

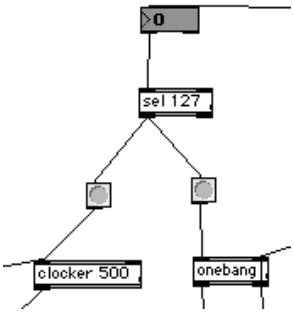


Figure 3. Max Patch onebang object

So each of the onebang objects were connected to a loadbang object so that they were ready at startup to let the first trigger in when the sensor output was equal to 127. Then it did nothing more until it received another bang in its right input inlet, which reset it. The left output of the onebang object was connected to a message box: 1. This message was passed on to the number box below it, that in turn triggered the soundfile playback object (sfplay~)(note: the tilda signifies MSP sound objects for Max/MSP, that is anything to do with sound). The sfplay~ object took a soundfile and a message in its left inlet (the message '1' told it to play the current sound file from beginning to end) and was connected to one or both of the channels of the audio output object (dac~) (the dac~ object is stereo by default).

The sound file must be opened before it's accessible to the sfplay~ object. Connected to the input of the sound file (and its 'open' message) was another loadbang object (Figure 4) so that the file would be accessible when the patch was opened. The same loadbang toggled the dac~ object on, and similarly we added a delay of 500ms to ensure reliability of the sequence.

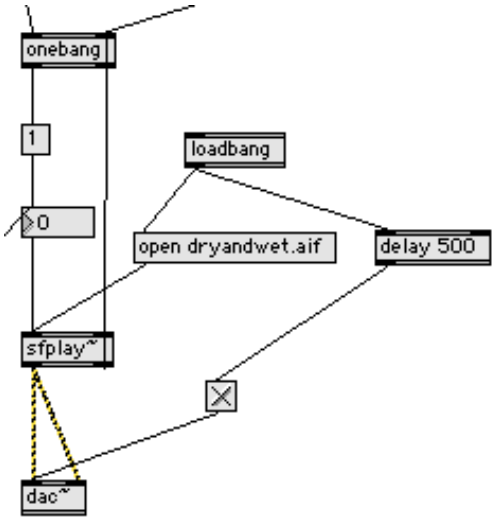


Figure 4. Max Patch onebang, sfplay~ and dac~ objects

As discussed, by using the onebang object we ensured that once the playback was triggered, it continued to play through without being reset each time the tilt switch was being toggled on and off by the swinging pendulum. But we still wanted to stop playback if the pendulum was no longer moving or if someone stopped the pendulum by choice. To do this we monitored the time elapsed between each cycle of the pendulum swing using the clocker object. The clocker object reports the time since it started, with an optional argument to set the output interval in milliseconds, in this case 500 (Figure 5).

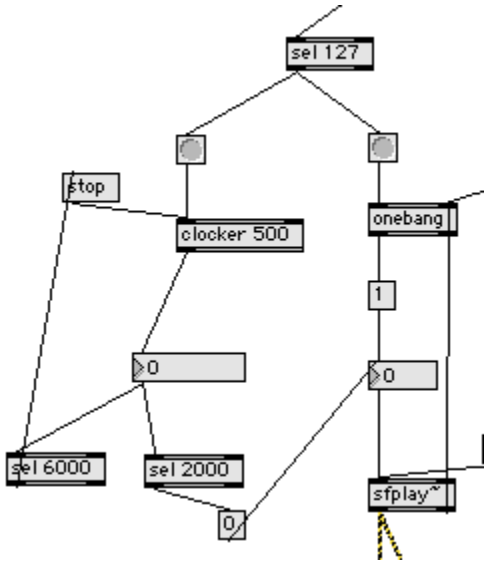


Figure 5. Max Patch clocker object

So from its left output, the select object had triggered the soundfile (onebang) and also started the metronome (clocker). The clocker object was restarted every time the tilt switch toggled on. The number box connected to the clocker's output monitored the time elapsed between each toggle of the switch in units of 500ms, and was in turn connected to two select objects. The first select object output a bang when the time reached 2000ms, the second select object output a bang when the time reached 6000ms (figure 5). If nothing had happened for 2 seconds, ie, the pendulum had stopped swinging for at least that long, the select object sent a bang to the message '0' which was passed on to the input of the sfplay~ object, and stopped playback. The right output of the sfplay~ object triggered a bang when it was done playing, which would happen if it had stopped (0) or if the audiofile had finished playing. That output was connected to the right inlet of the onebang object (figure 5), so it was ready to trigger if the pendulum was moving again. If nothing had happened for another 4 seconds (a total of 6000ms), the next select object sent a message to the clocker to stop monitoring, and the whole patch was idle until a pendulum started swinging. (Kershaw 2011; Castaldi 2011).

The Max/MSP patch we are developing for the Wild Weather installation will work along similar lines to the one we developed for the Memory Flows exhibition but somewhat in reverse where the audio files sent to the patch and cued via FTP will trigger the swinging movement of the resonators. As mentioned, this movement could be triggered by multiple media file upload via FTP and will use 5 actuators located at the top of each of the five resonators to control their back and forth swinging motion. Up to 5 audio files will be cued for playback and previously uploaded files will be looped for random playback. The proposed sound (natural or otherwise) inside the resonators will be fed back into the computer, processed in Max/MSP and amplified back through the resonator's speaker system in real time.

CONCLUSION

This installation sits at the intersection of sonic art, networked media, human ecology and remote user interaction. Through the combined technologies of the Internet, a computer/sensor interface and a public sound sculpture the project will investigate shared experiences and provide a means for the general population to contribute meaningful content that addresses our intense and dangerous relationship with the weather for public and web broadcast. The weather is a contentious and emotionally charged topic for many of us and over time I hope that the work will generate a rich pool of personal response to the global issues of climate change and global warming with the possibility of attracting further attention (from the public and media) and acting as a catalyst for change to the ever increasing dangers of wild weather.

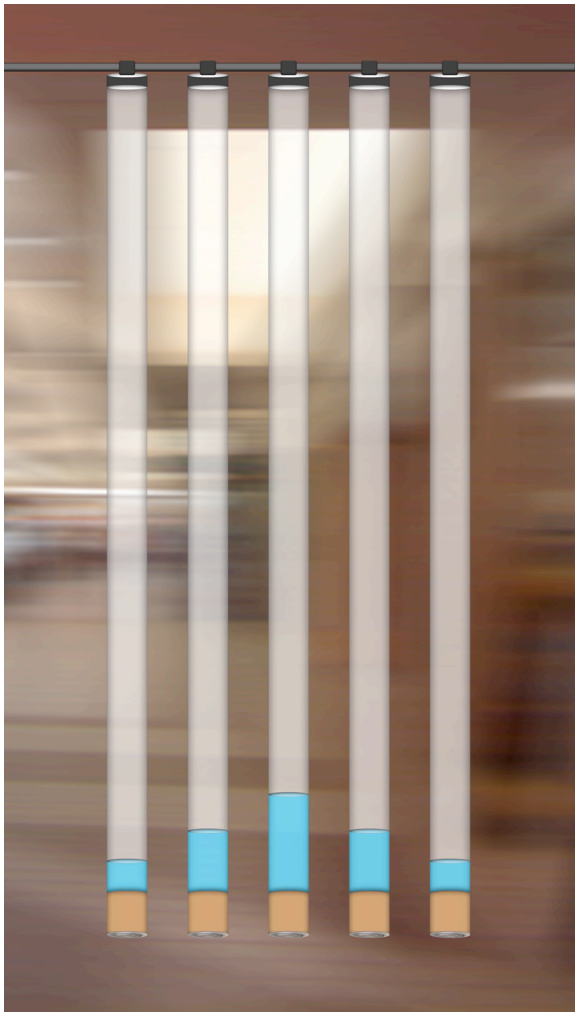


Figure 6. Wild Weather sound installation prototype

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